

CHM111 Practice Exam 2 on chapter(s): 3, 4

1. 2.7%
2. $\text{Fe}_2\text{O}_3 + 3 \text{CO} \rightarrow 2 \text{Fe} + 3 \text{CO}_2$
3. 2.56 g
4. 34.1 mL
5. 0.758 mol
6. 16.11 g
7. AB_2
8. 51.8%
9. 3.06 mL
10. 18.7 g
11. $\text{CH}_3\text{OH} + \text{CO} \rightarrow \text{CH}_3\text{COOH}$
12. Oxygen
13. 0.638 moles
14. 2.36 M
15. $2.06 \times 10^1 \text{ g}$
16. $17 \text{O}_2 + 2 \text{C}_6\text{H}_{12}\text{O} \rightarrow 12 \text{CO}_2 + 12 \text{H}_2\text{O}$
17. 95%
18. 0.459 M
19. SO
20. $\text{C}_7\text{H}_{14}\text{O}$
21. 0.0134 moles
22. 49.7%
23. +3
24. neither oxidized nor reduced
25. 128 g/mol
26. $8.148 \times 10^{49} \text{ g}$
27. 16%
28. 7.44×10^{22} fluorine atoms
29. 32.3 L

30. 7.3 mM

31. 17.0 g

32. $\text{S}_8 + 12 \text{O}_2 + 8 \text{H}_2\text{O} \rightarrow 8 \text{H}_2\text{SO}_4$

33. $381.4 \frac{\text{g}}{\text{mol}}$

34. 0.85339 mol

35. 3.96 g

36. Single-Displacement

37. 40.30 g

38. 32%

39. $\text{TiO}_2 + 2 \text{Cl}_2 + \text{C} + 2 \text{Mg} \rightarrow \text{Ti} + \text{CO}_2 + 2 \text{MgCl}_2$

40. ionic, base